EXPERIMENT:06

Write a C program for ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is “B,” and the second most frequent letter of the ciphertext is “U.” Break this code.

Code:

```
#include <stdio.h>
int main()
{
   char cipher[200];
   int a, b, i, c, p;
   printf("Enter the ciphertext: ");
   fgets(cipher, sizeof(cipher), stdin);
   // B = 1, E = 4
   // U = 20, T = 19
   // 15a = 19 (mod 26)
   // a = 3
   a = 3;
   // 4a + b = 1 (mod 26)
   // b = 15
   b = 15;
   printf("Affine key: a = %d, b = %d\n", a, b);
   printf("Decrypted text: ");
   // Inverse of 3 modulo 26 is 9
   for(i = 0; cipher[i] != '\0'; i++)
   {
      if(cipher[i] >= 'A' && cipher[i] <= 'Z')
      {
         c = cipher[i] - 'A';
         p = (9 * (c - b)) % 26;
         if(p < 0)
            p = p + 26;
```

```
            printf("%c", p + 'A');
        }
        else
        {
            printf("%c", cipher[i]);
        }
    }
    return 0;
}
```

OUTPUT:

